

Name: \_\_\_\_\_

Registration: \_\_\_\_\_

Design an 8x1 Mux using as building blocks the 2x1 and 4x1 Muxes shown below. Use the provided skeleton file to implement the 8x1 Mux.

(Simulation: 50% pts., FPGA Implementation: 50% pts.)

```
module m21 (OUT, A, B, S);  
    input A, B, S;  
    output OUT;  
  
    assign OUT=S?B:A;  
endmodule
```

```
module m41 (OUT, A, B, C, D, S1, S0);  
    input A, B, C, D, S1, S0;  
    output OUT;  
    reg OUT;  
    always @(*)  
        case ({S1, S0})  
            0: OUT=A;  
            1: OUT=B;  
            2: OUT=C;  
            3: OUT=D;  
        endcase  
endmodule
```

```
module m81 (OUT, I, Sel);  
    input [7:0] I;  
    input [2:0] Sel;  
    output OUT;  
    //Write your code below
```

```
endmodule
```

```

module m81 (OUT, I, Sel);
    input [7:0] I;
    input [2:0] Sel;
    output OUT;
    //Write your code below
    wire O1, O2;
    m41 m1 (O1, I[0], I[1], I[2], I[3], Sel[1], Sel[0]);
    m41 m2 (O2, I[4], I[5], I[6], I[7], Sel[1], Sel[0]);
    m21 m3 (OUT, O1, O2, Sel[2]);
endmodule

```

```

module test_m81;
    reg [7:0] I;
    reg [2:0] Sel;
    wire OUT;

    m81 dut (OUT, I, Sel);

    initial begin
        I=8'b10110101;
        #10 Sel=3'd0;
        #10 Sel=3'd0;
        #10 Sel=3'd1;
        #10 Sel=3'd2;
        #10 Sel=3'd3;
        #10 Sel=3'd4;
        #10 Sel=3'd5;
        #10 Sel=3'd6;
        #10 Sel=3'd7;

        end
endmodule

```

